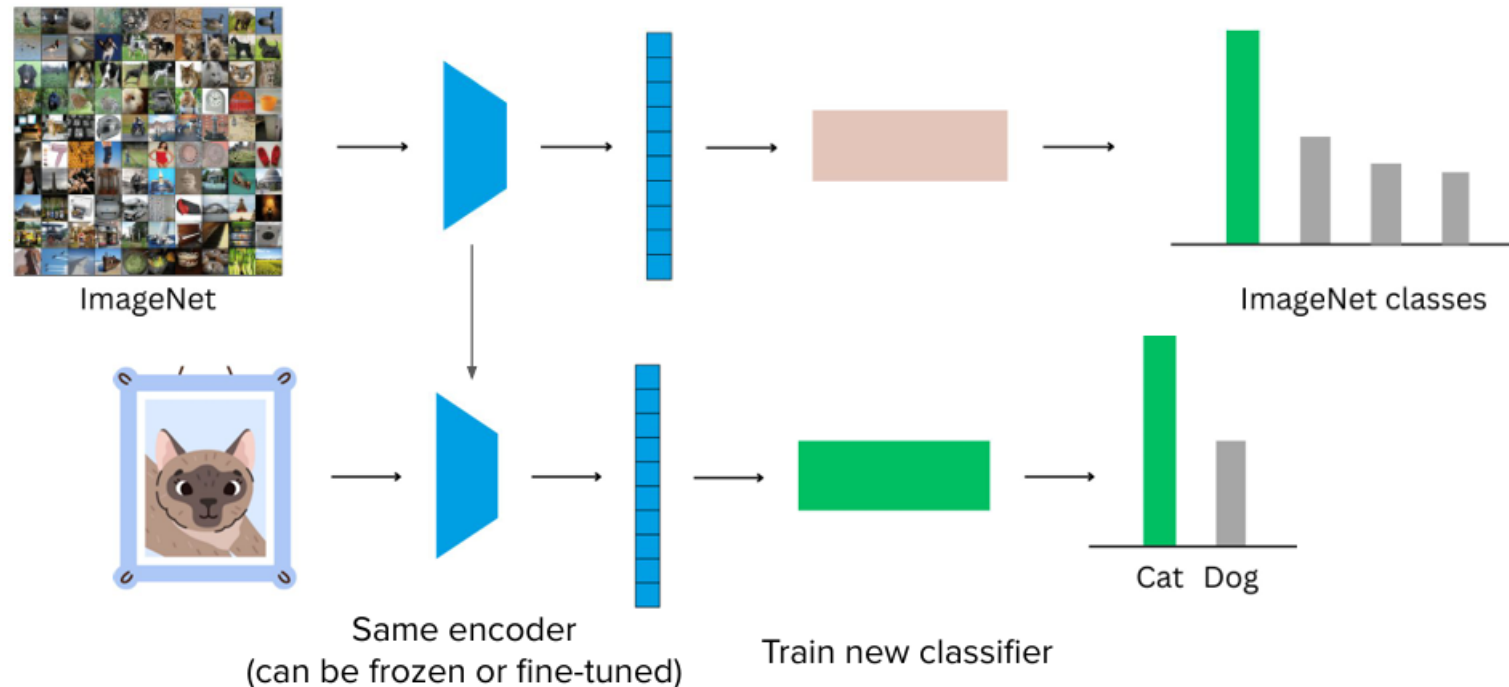


5 - Self-Supervised Learning

Contrastive Learning, Dimensionality Reduction, Face Recognition, GenAI

Self-Supervised Learning

- If you want to solve a computer vision problem:
 - <2014: Handcraft features for each task + ML classifiers.
 - 2014-2020: Train supervised models on large labeled datasets, fine-tune for your task.
- **Transfer Learning:** transfer representations to different tasks.

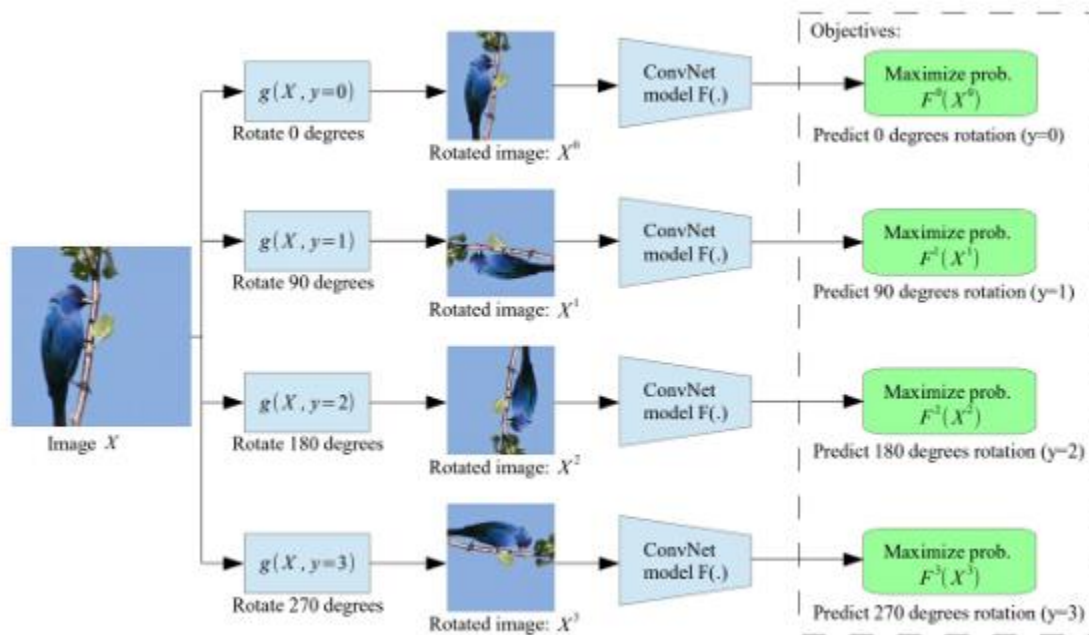


Self-Supervised Learning

- You cannot label *all* data on the internet, but to make deeper models you need more data one way or another...
- **Self-Supervised Learning (SSL)** doesn't need labeled data:
 1. Pre-train an encoder to solve *auxiliary tasks* using annotations that can be created in a semi-automated way.
 2. Fine-tune for your task using supervised learning.
- If the pre-trained model is trained well, only a limited amount of labeled data is needed for fine-tuning.

Self-Supervised Learning - Transformations

- The first SSL models created learning tasks from data without annotations.
- **RotNet**: predict rotation angles (*transformations*) applied to images.
- Fine-tune to transfer its learned representations to other tasks.

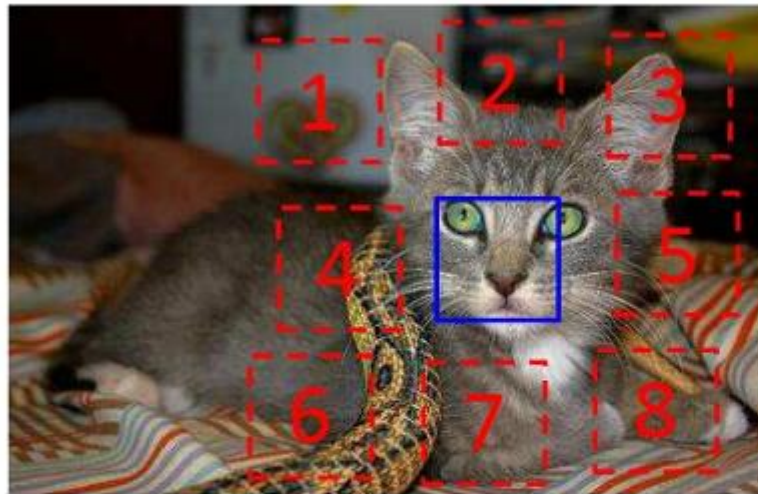


# Rotations	Rotations	CIFAR-10 Classification Accuracy
4	0°, 90°, 180°, 270°	89.06
8	0°, 45°, 90°, 135°, 180°, 225°, 270°, 315°	88.51
2	0°, 180°	87.46
2	90°, 270°	85.52

Intuition: if a model recognizes transformations, it has ‘*visual commonsense*’: it knows *something* about the object and how it should look without a transformation.

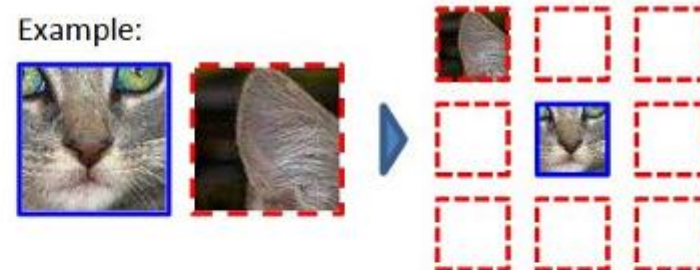
Self-Supervised Learning - Context Prediction

- The first SSL models created learning tasks from data without annotations.
- **Context Prediction:** predict *relative* position of two image patches.



$$X = (\text{cat face patch}, \text{cat ear patch}); Y = 3$$

Example:



Question 1:

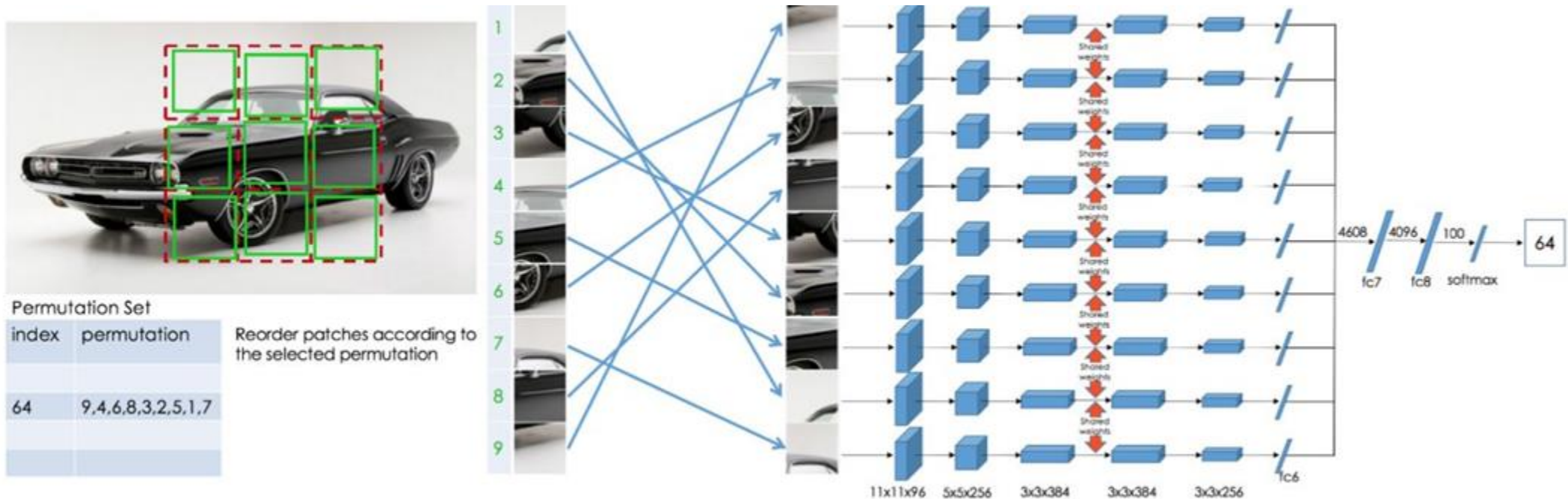


Question 2:



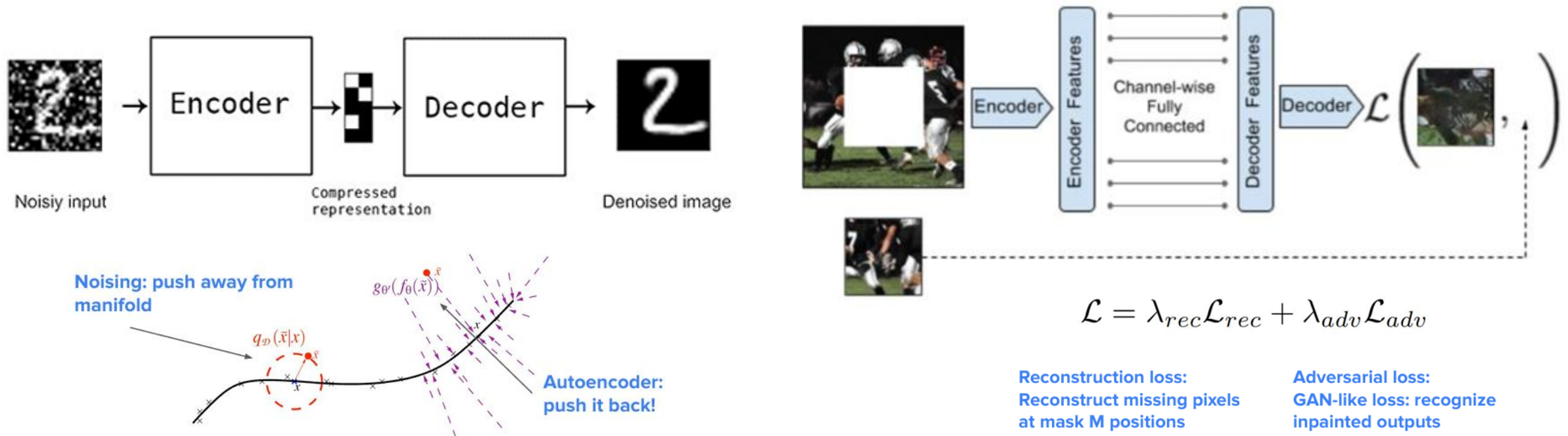
Self-Supervised Learning - Permutations

- The first SSL models created learning tasks from data without annotations.
- **Image Permutations:** make the model solve a '*jigsaw puzzle*'.



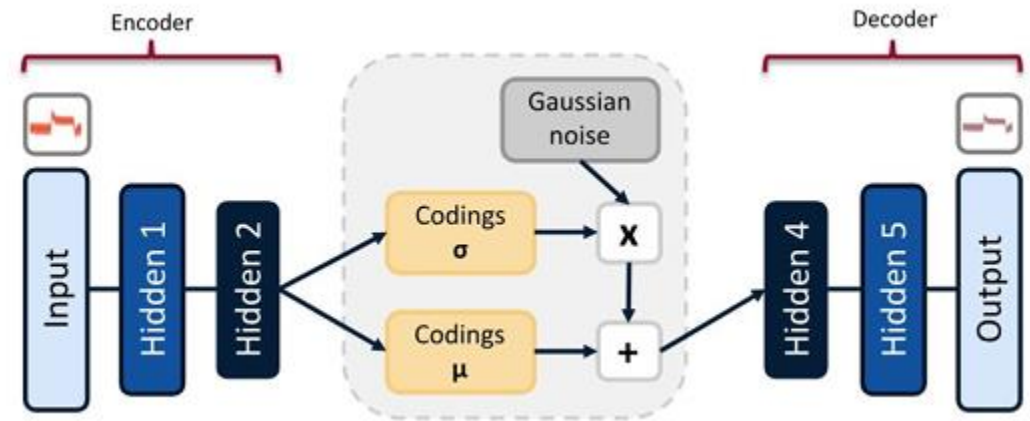
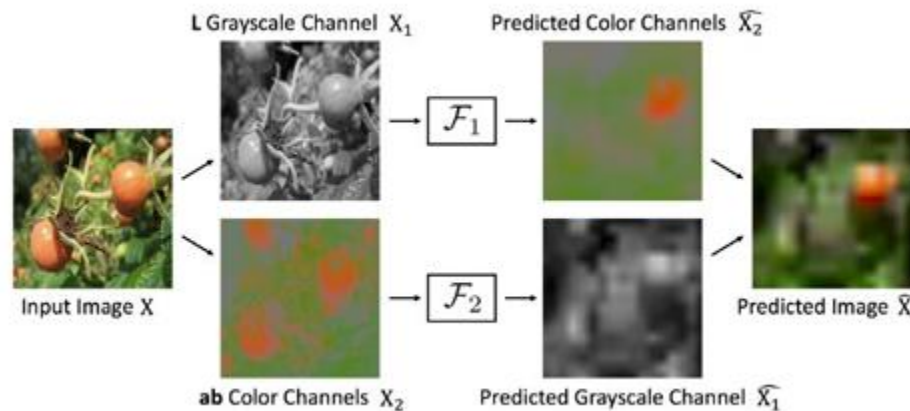
Self-Supervised Learning - Reconstruction

- The first SSL models created learning tasks from data without annotations.
- **Denoising Autoencoder:** learn to reconstruct the input.
- **Inpainting Autoencoder:** learn to reconstruct missing segments.



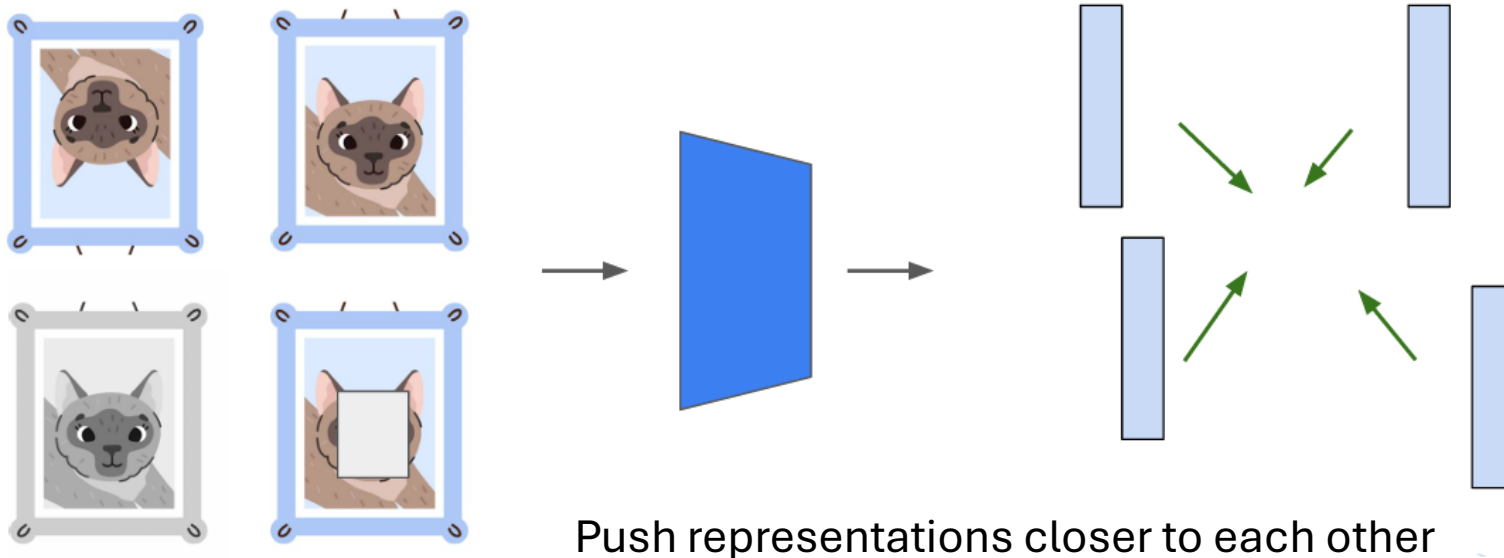
Self-Supervised Learning - Reconstruction

- The first SSL models created learning tasks from data without annotations.
- **Split-Brain Autoencoder**: learn to predict image channels.
- **Variational Autoencoder (VAE)**: learn to model latent representations as probability distributions (enables generative capabilities).



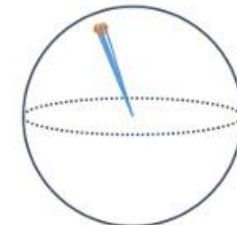
Contrastive Learning

- Models can become experts in solving these tasks, but the learned representations still lack *generalization*.
- Can we learn representations that group similar images in the feature space?
- No *annotations*, but we can use different *augmentations* of the same image.

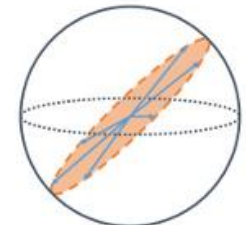


Possible issues:

1. **Complete Collapse:** producing same representation for any input.
2. **Dimension Collapse:** unused dimensions in representations.



(b) complete collapse



(c) dimensional collapse

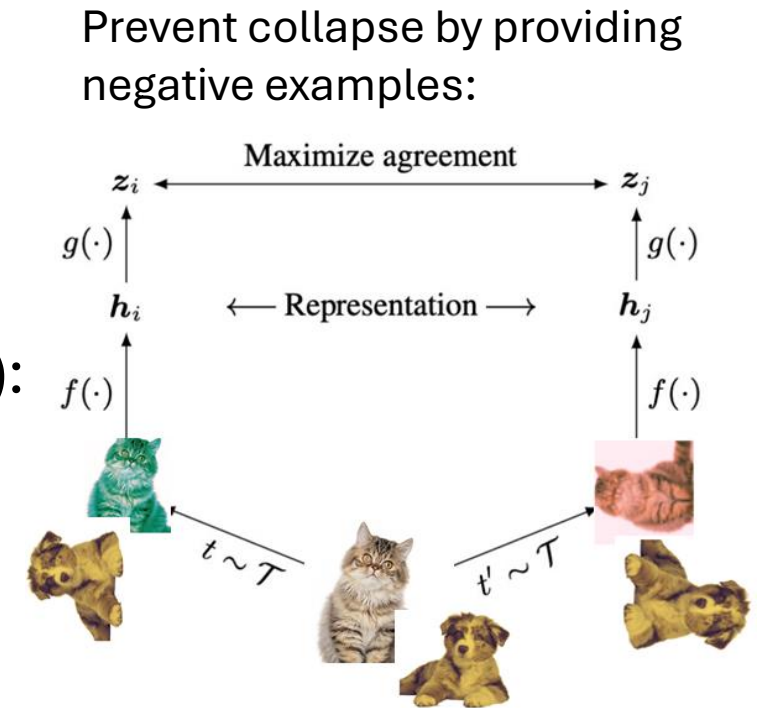
Contrastive Learning

- **Simple Framework for Contrastive Learning (SimCLR):**

1. Take a batch of N images.
2. Apply augmentations to each image.
3. Pass augmented images through encoder f (e.g. CNN).
4. Pass representations through project head g (e.g. MLP).
5. Compute *contrastive loss* between projected representations.

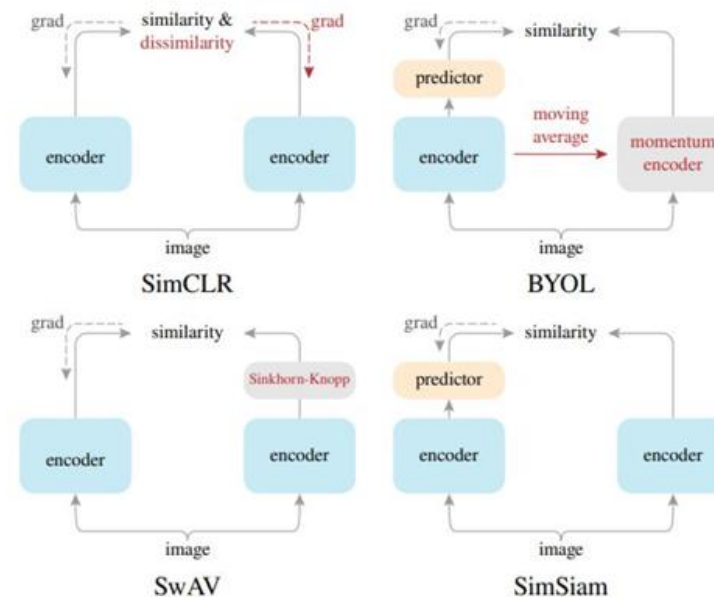
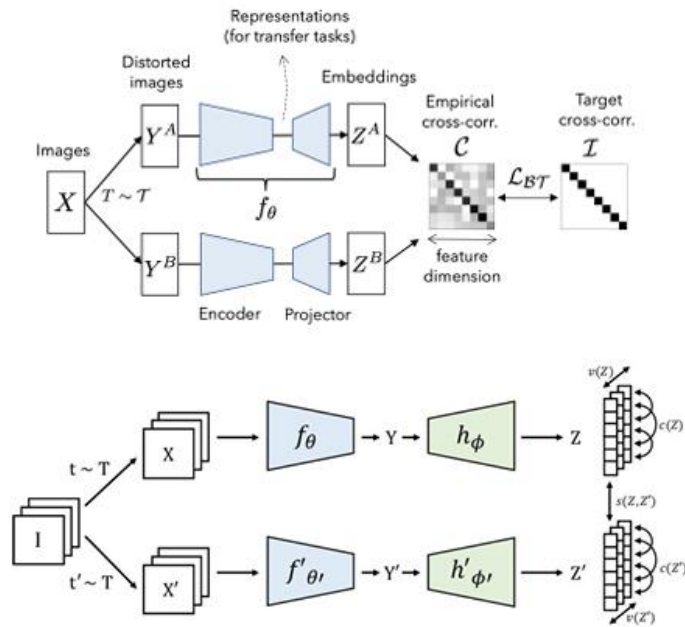
- **Contrastive Loss:** maximize similarity between *positive pairs* (augmented views of the same images), minimize agreement between *negative pairs*.

- Outperforms supervised models when annotations are limited.
- Requires large batch sizes (N) to provide enough negative examples.



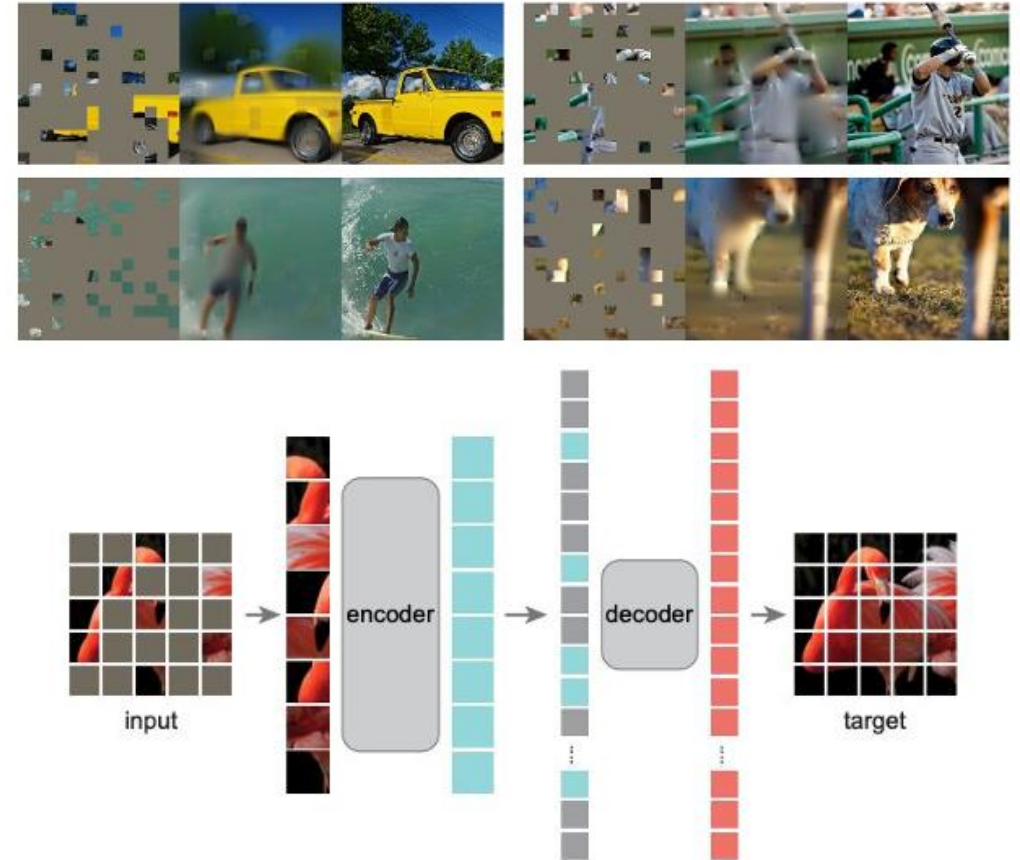
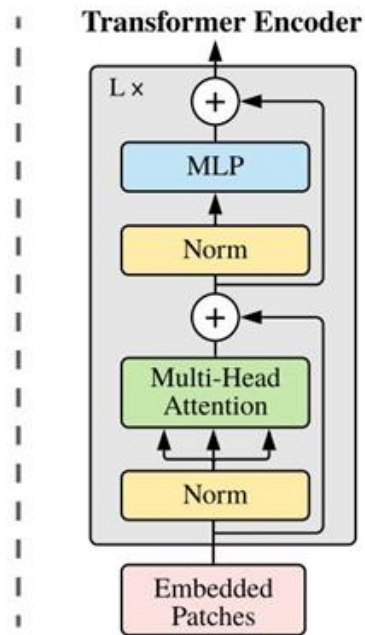
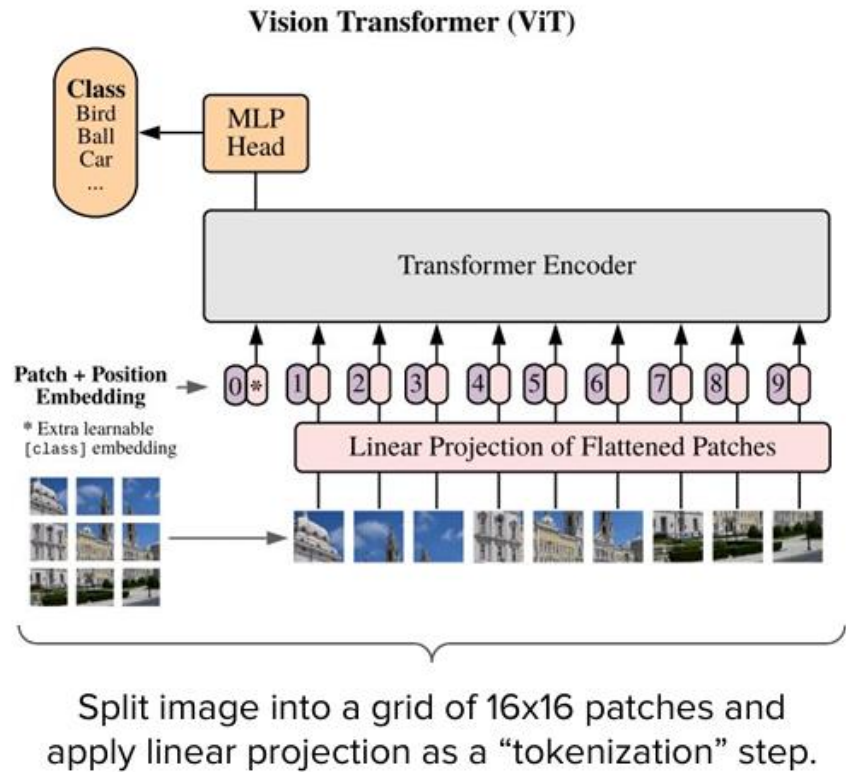
Contrastive Learning

- *Barlow Twins, VICReg, MoCo, BYOL, SWaV, SimSiam, ...* → other approaches that try to address the challenges of contrastive learning.
- **Non-Contrastive Learning:** maximize similarity of positive pairs *without* using negative pairs (no need for larger batch sizes), prevent collapse through some form of regularization (e.g. *cross-correlation regularization*).



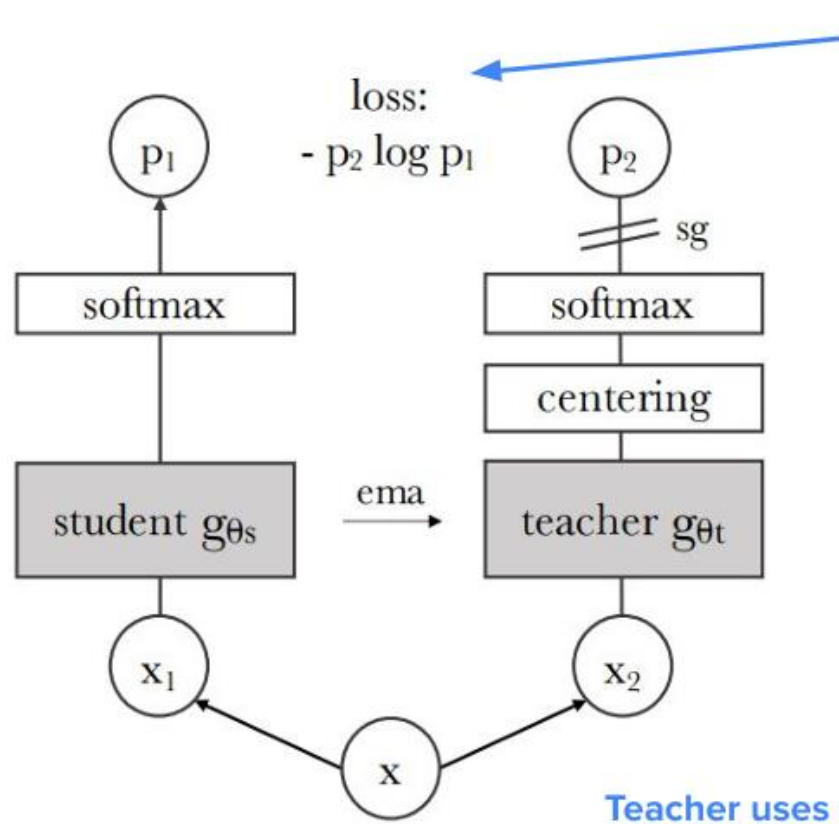
Self-Supervised Learning - ViTs

- **Masked Autoencoder:** reconstruct masked patches from unmasked ones.



Self-Supervised Learning - ViTs

- **DINO**: combining joint-embedding architecture (contrastive learning) with ViTs.



New loss
(inspired by distillation)

Softmax over output (feature) dimensions:

$$P_s(x)^{(i)} = \frac{\exp(g_{\theta_s}(x)^{(i)} / \tau_s)}{\sum_{k=1}^K \exp(g_{\theta_s}(x)^{(k)} / \tau_s)},$$

Minimize **cross-entropy** between teacher and student feature probabilities:

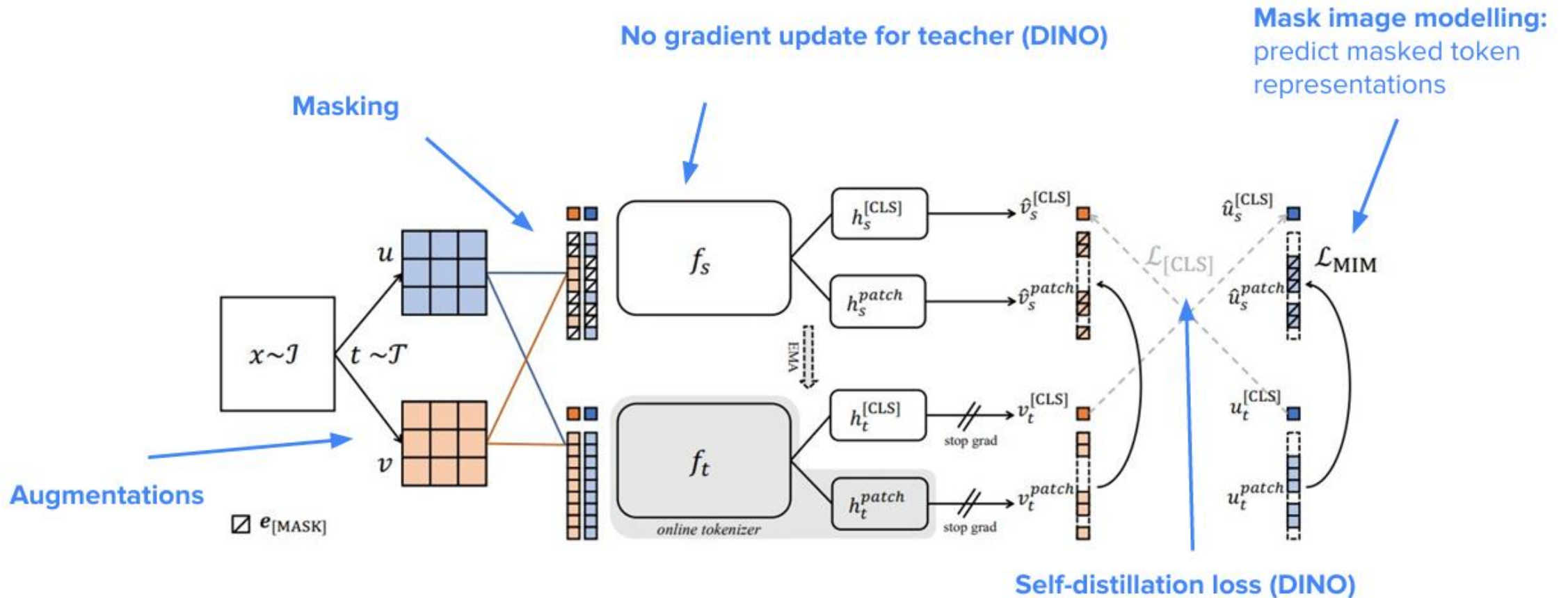
$$\min_{\theta_s} \sum_{x \in \{x_1^g, x_2^g\}} \sum_{\substack{x' \in V \\ x' \neq x}} H(P_t(x), P_s(x'))$$

Teacher uses global views

Student – local image views

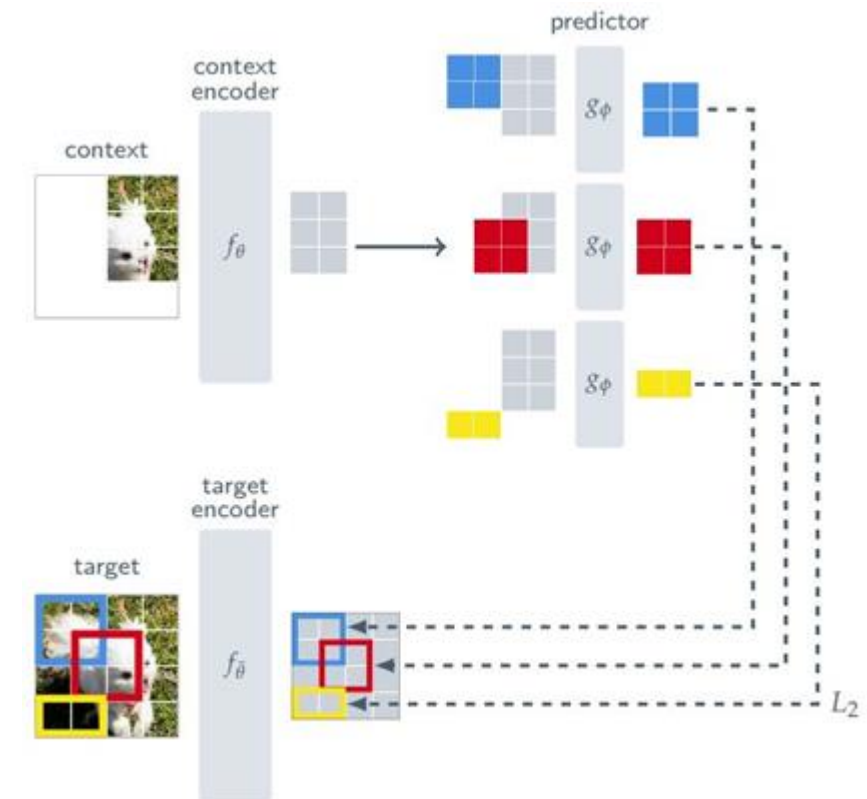
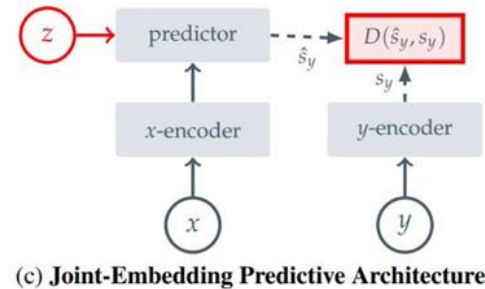
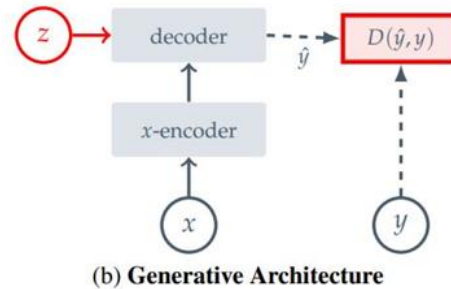
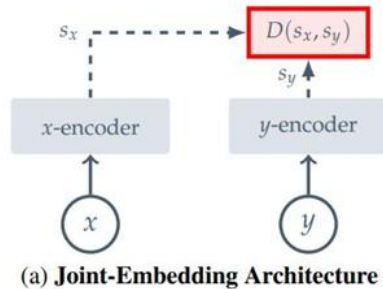
Self-Supervised Learning - ViTs

- **iBot**: combining DINO with masked autoencoders.



Self-Supervised Learning - JEPA

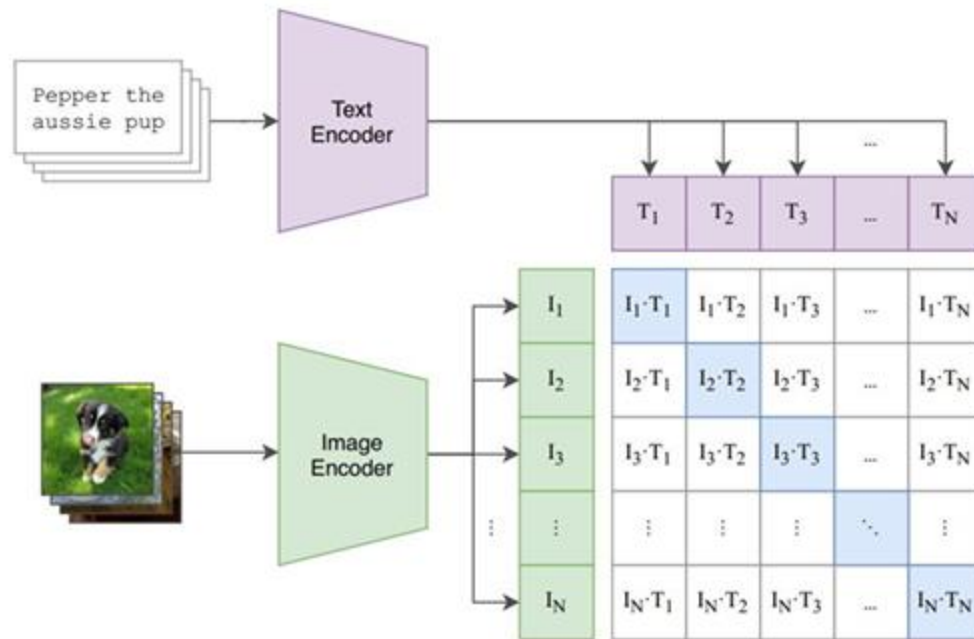
- Whereas previous autoencoders make predictions in the image space and ViTs are generative, **Joint Embedding Predictive Architecture (JEPA)**, learns to reconstruct images in the *representation* space.



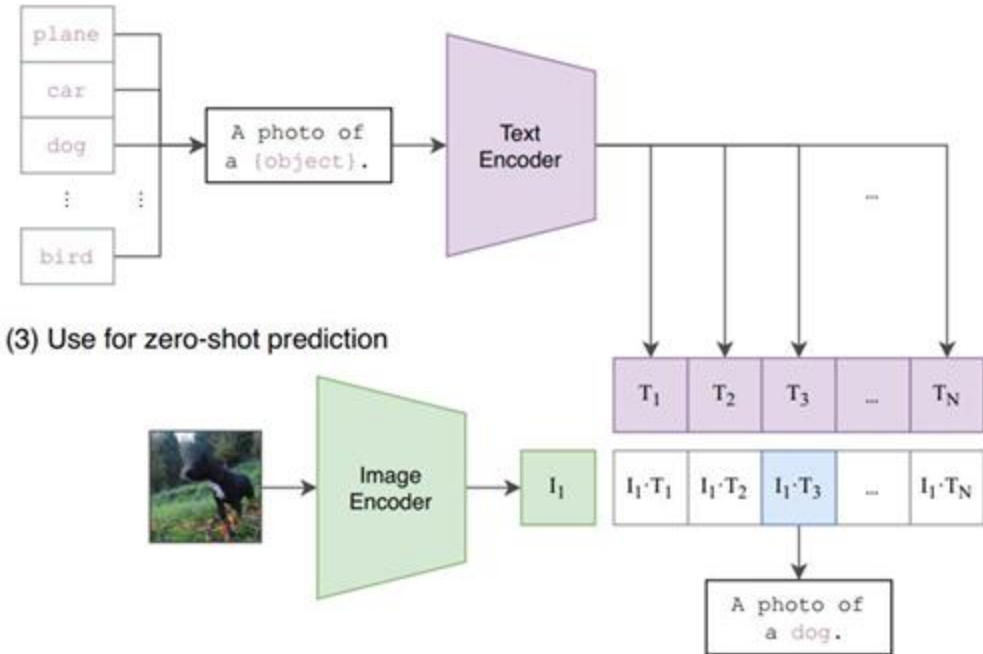
Self-Supervised Learning - CLIP

- **CLIP**: learns a joint representation for images and text using contrastive loss.

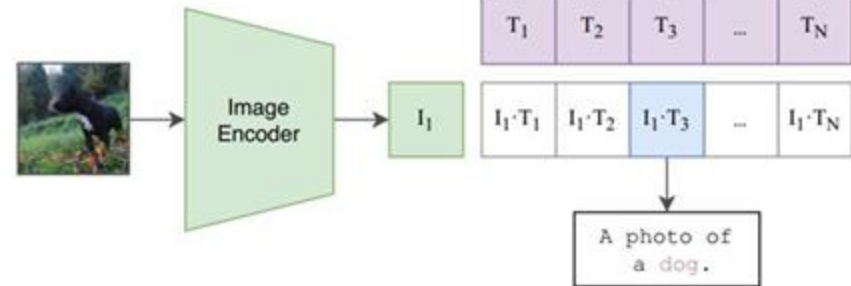
(1) Contrastive pre-training



(2) Create dataset classifier from label text

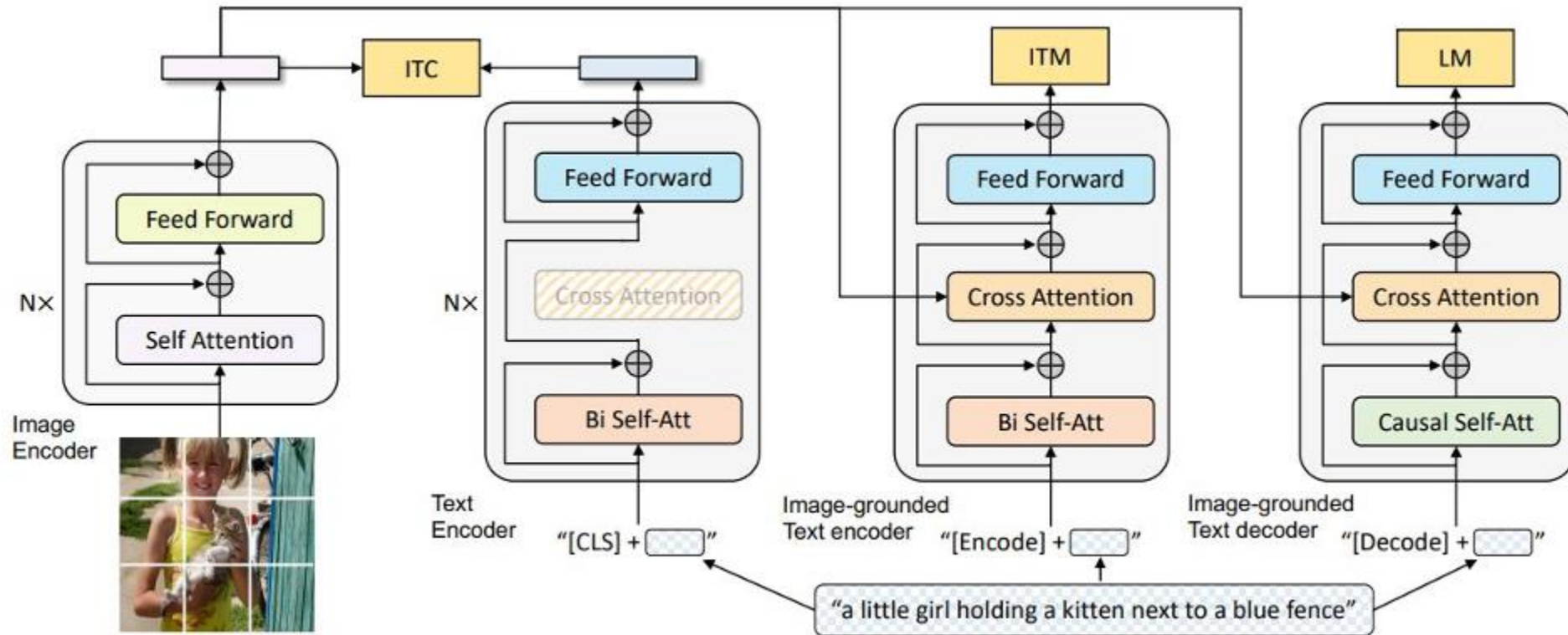


(3) Use for zero-shot prediction



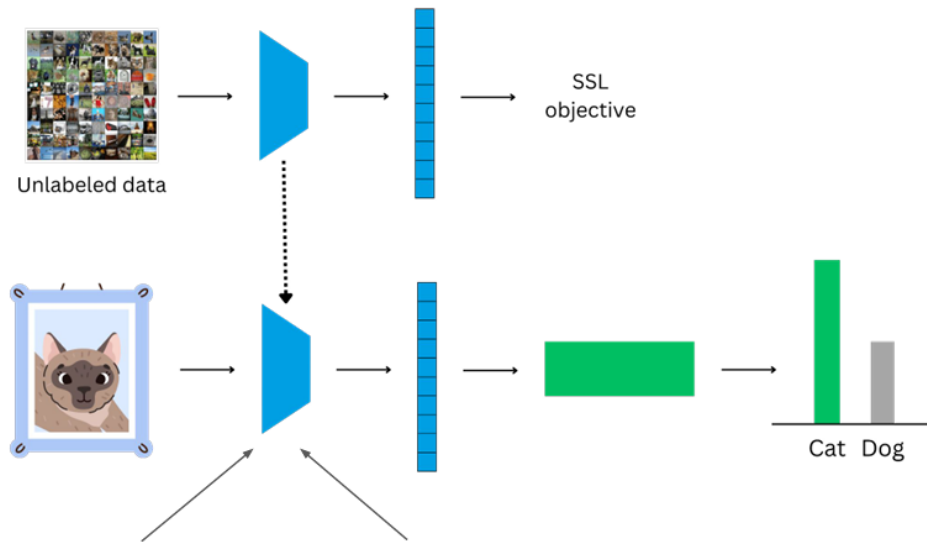
Self-Supervised Learning - BLIP

- **BLIP**: jointly trains an image encoder, image-grounded text encoder, and decoder (enables tasks like visual question answering).



Self-Supervised Learning - Fine-tuning

- How to reach better performance at lower cost?
 - **Partial Fine-tuning**: only unfreeze some layers in the encoder.
 - **Prompt Tuning**: add learnable tokens that are only learned during fine-tuning.
 - **Low-Rank Adaptation (LoRA)**: ingest trainable weight matrices with low ranks.

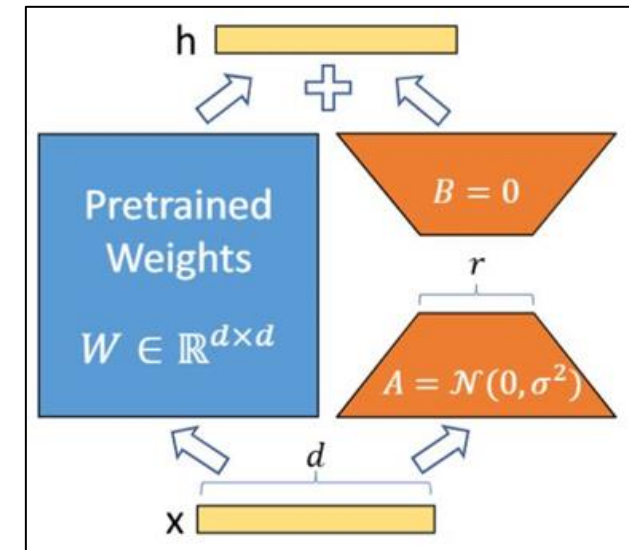
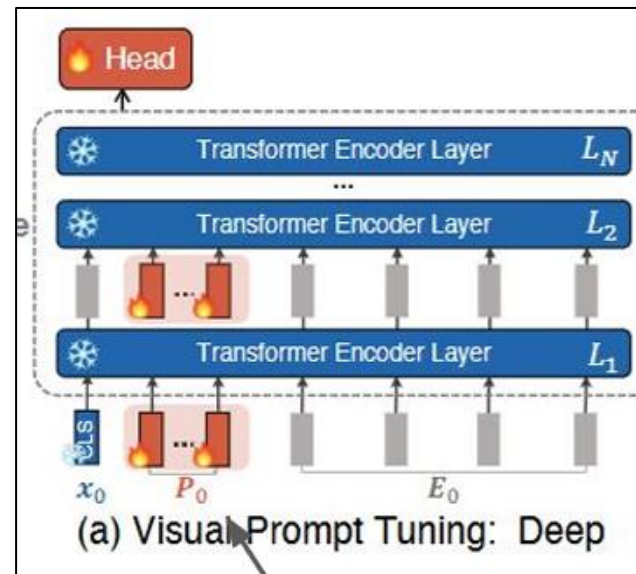


Frozen encoder:

- + Lightweight – only classifier is tuned
- Suboptimal performance (esp. if domains are too different)

Fine-tuning encoder as well:

- + Predictive performance
- Needs a lot compute
- Needs a lot of data

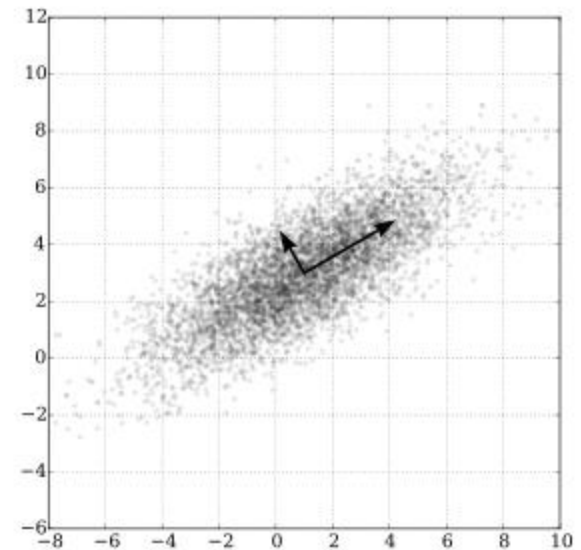


2 smaller matrices instead of 1 big

Dimensionality Reduction - PCA

- Given: N data points $x_1, \dots, x_N \in \mathbb{R}^d$ (normalized with mean 0, variance 1)
- The **principal components** of the data are equal to the eigenvectors of the data's covariance matrix.

$$\begin{pmatrix} \mathbf{w}_1 \\ \mathbf{w}_2 \\ \vdots \\ \mathbf{w}_d \end{pmatrix} = \begin{pmatrix} \mathbf{u}_1^T \\ \mathbf{u}_2^T \\ \vdots \\ \mathbf{u}_d^T \end{pmatrix} (\mathbf{x} - \boldsymbol{\mu}) \Leftrightarrow \mathbf{W} = \mathbf{U}(\mathbf{x} - \boldsymbol{\mu})$$

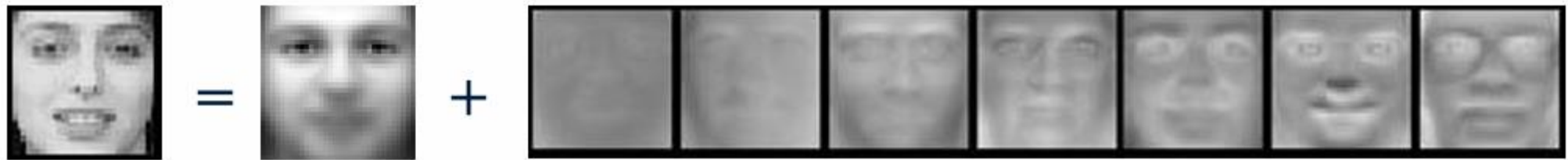


Dimensionality Reduction - Autoencoders

- By constraining a network to learn a compressed representation, redundant or less important information is eliminated.
- **Denoising Autoencoders** are great at providing robust and non-overfitted representations.
- **Sparsity Constraint:** only a few hidden units are allowed to activate at once.
- **Hierarchical Autoencoder:** organizes latent variables into multiple layers to capture features at different scales.
 - Encoder: bottom-up (each subsequent layer becomes more abstract)
 - Decoder: top-down (each subsequent layer becomes less abstract)

Face Recognition - Eigenfaces

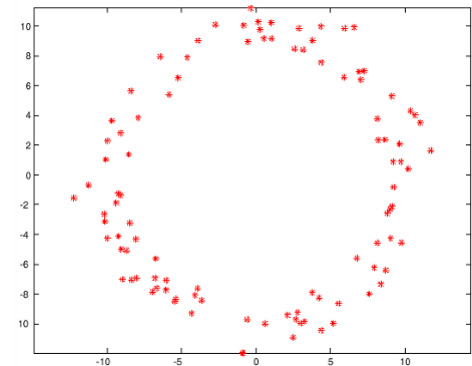
- The image space is extremely high-dimensional, but images of faces have a lot in common and should lie somewhere close to each other in this space.
- **Eigenfaces:** use PCA to determine the vectors (*'eigenfaces'*) that span the low-dimensional subspace of all faces, represent each face as a linear combination of eigenfaces.



$$\hat{\mathbf{x}} = \boldsymbol{\mu} + w_1 \mathbf{u}_1 + w_2 \mathbf{u}_2 + \dots + w_k \mathbf{u}_k$$

Face Recognition - Eigenfaces

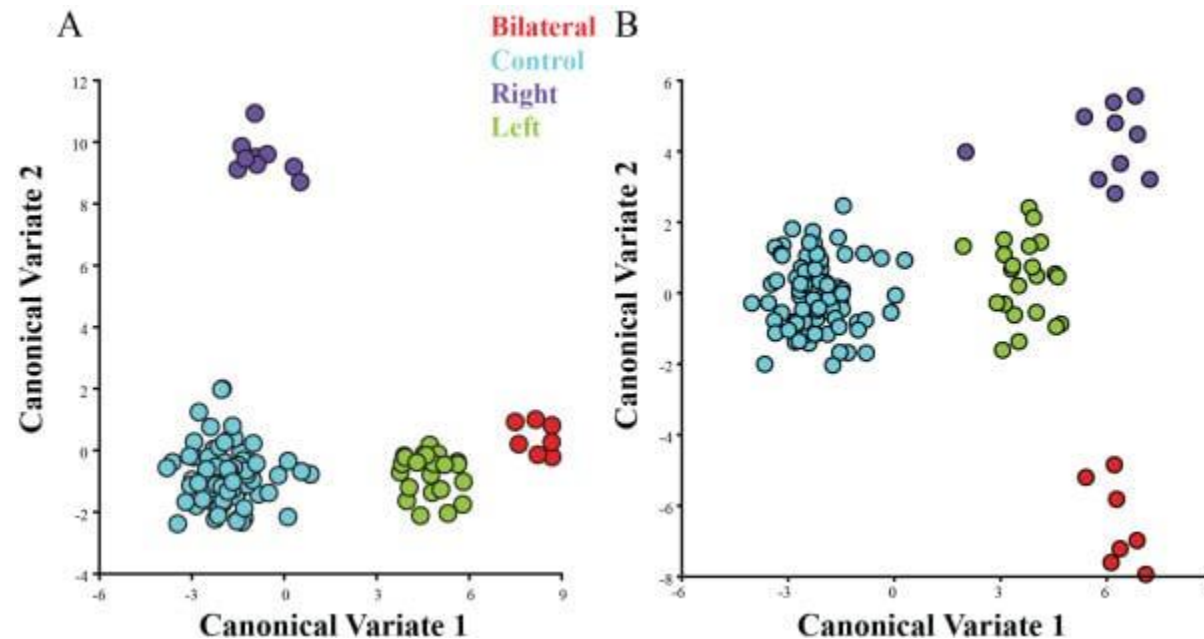
- Training:
 - Find mean face μ and covariance matrix Σ of the face images.
 - Find k principal components / eigenfaces $u_1, \dots, u_k$ (eigenvectors of Σ).
 - Project each image x_i onto the subspace: $(w_{i1}, \dots, w_{ik}) = (u_1^T(x_i - \mu), \dots, u_k^T(x_i - \mu))$.
- Recognition:
 - Project new face x onto the subspace.
 - *(Optional) reconstruct x , check the reconstruction error $x - \hat{x}$ to verify if x is a face.*
 - Classify x as the closest training face in the subspace.
- Limitations:
 - Not robust to misalignment and background variations.
 - Projection may suppress important details.



This dataset cannot be described well using PCA.

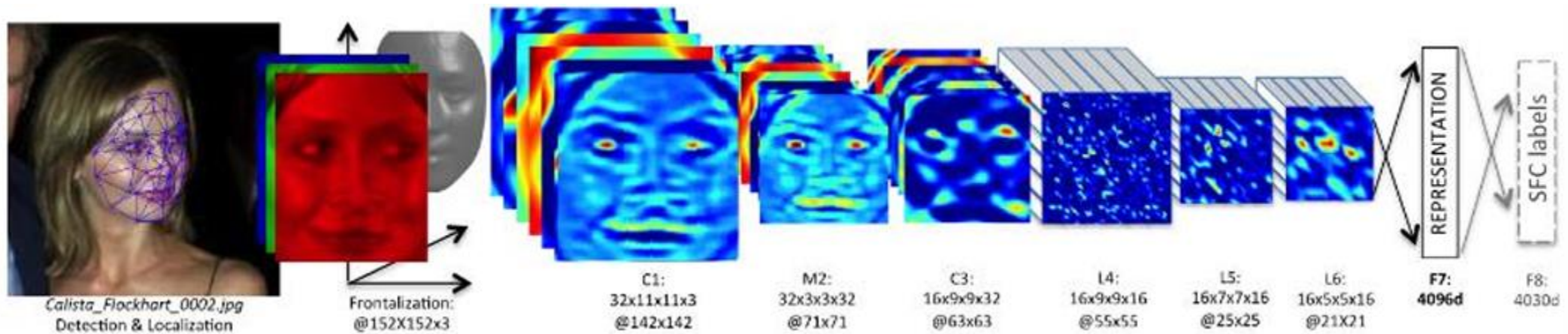
Face Recognition - Canonical Variates

- **Canonical Variates:** choose linear combinations of the features that allow for good distinction between classes (maximize spread of class means).



Face Recognition - DeepFace

- Uses CNNs to transform faces into 4096-dimensional vectors
- It first detects features of a face (center of eyes, tip of nose, etc.) and uses **frontalization** to translate the face onto a warped image.
- It achieves 97% accuracy on face recognition, better than most humans.

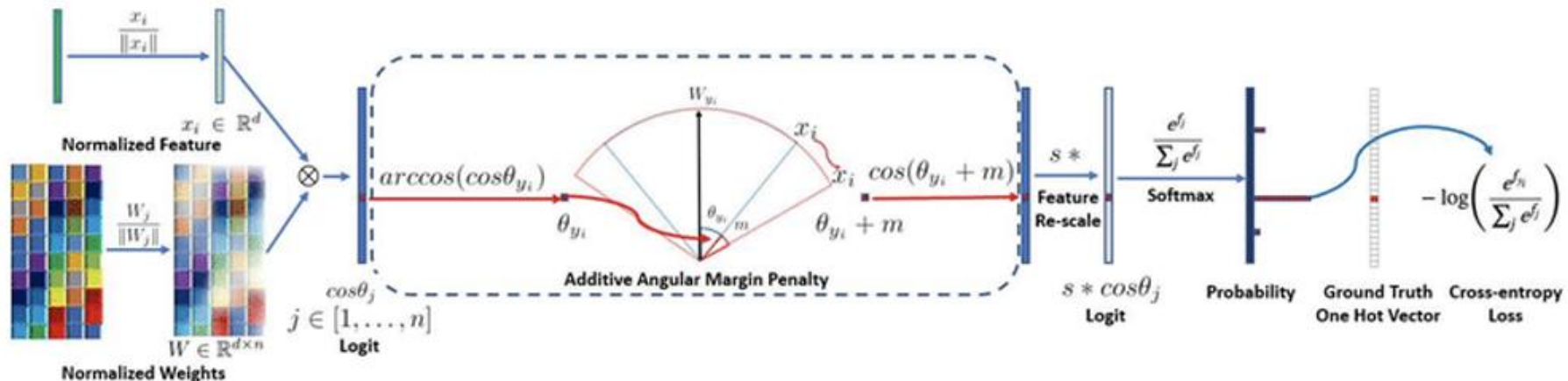


Face Recognition - InsightFace

- It achieves 99.8% accuracy on face recognition.
- Main innovation is a new loss function, **ArcFace Angular Loss**:

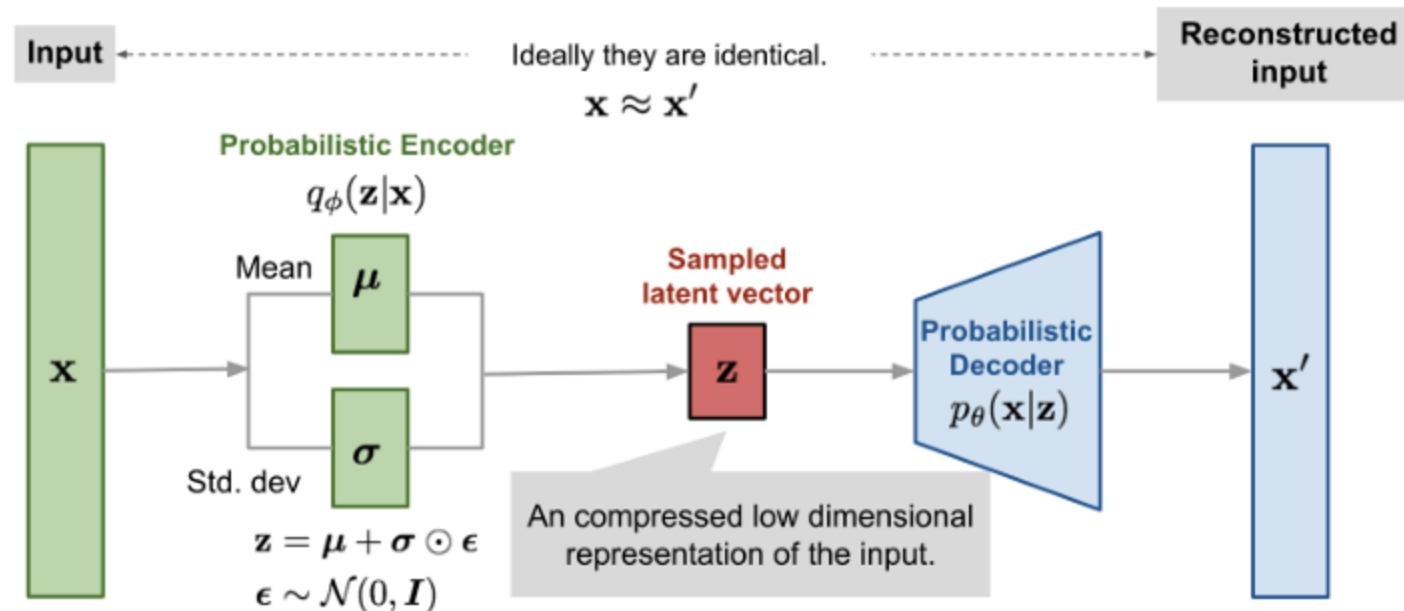
$$L = -\frac{1}{N} \sum_{i=1}^N \log \frac{e^{s \cdot \cos(\theta_{y_i} + m)}}{e^{s \cdot \cos(\theta_{y_i} + m)} + \sum_{j \neq y_i} e^{s \cdot \cos(\theta_j)}}$$

- Where θ_{y_i} is the angle between the feature vector and the correct class center, m is the additive angular margin (constant ~ 0.5), s is the scaling factor (usually 64), and N is the batch size.



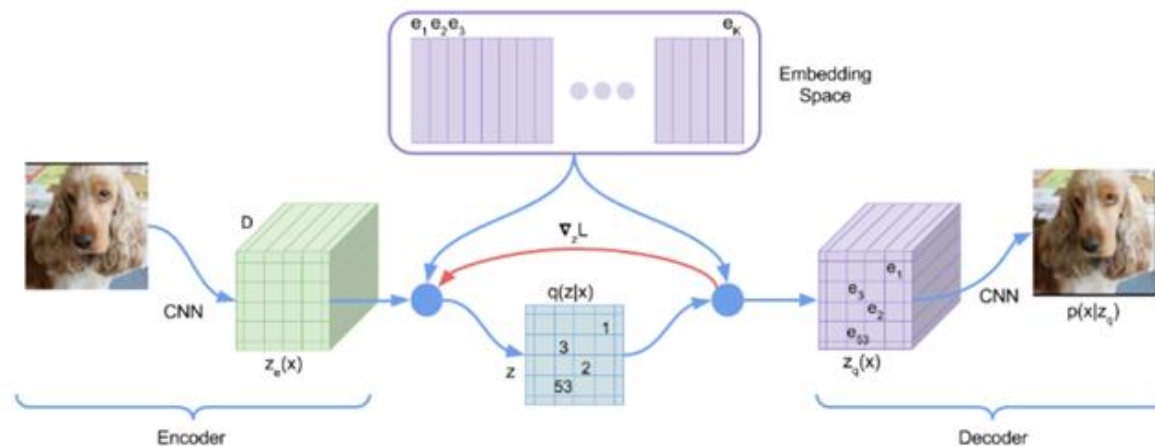
Generative AI - VAEs

- VAEs are ideal for generative modeling because of their continuous latent space, which can easily be sampled and interpolated from.



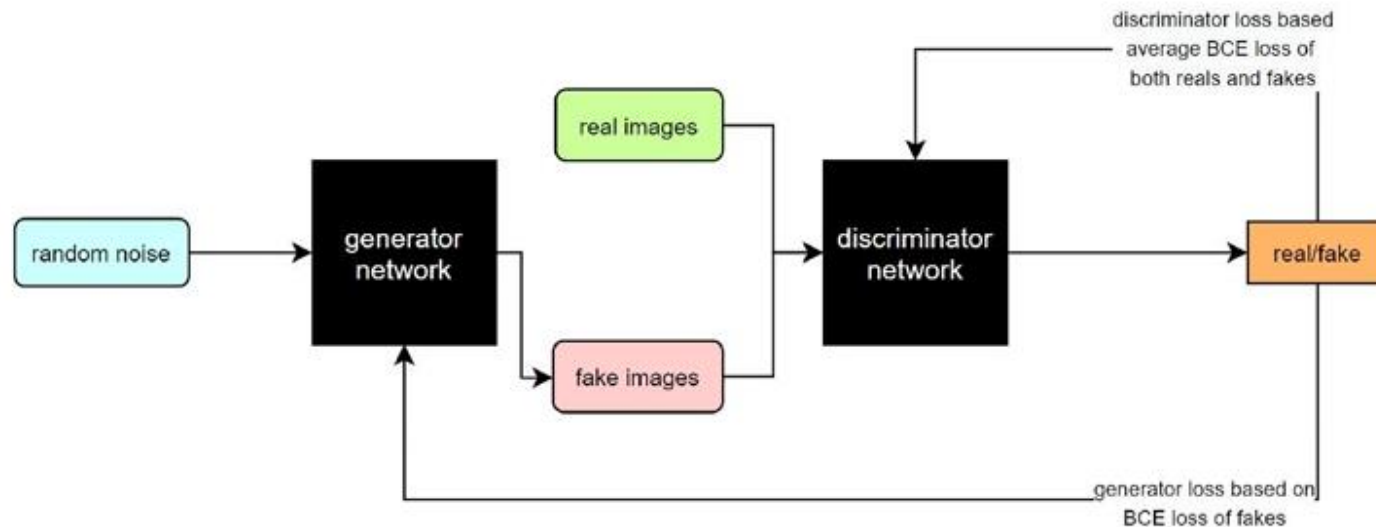
Generative AI - VAEs

- **VQ-VAE** takes the continuous latent output of the encoder and ‘snaps’ it to the nearest vector in a learned codebook (discrete latent space).
- This helps prevent *posterior collapse* (all inputs map to the same vector), and produces sharper, more detailed images.
- Because a rounding operation is non-differentiable, gradients pass unchanged from the decoder to the encoder, while the codebook vectors are adjusted through a separate loss function.



Generative AI - GANs

- **Generator:** generates fake images from random noise (single forward pass).
- **Discriminator:** predicts whether an image is real or fake.

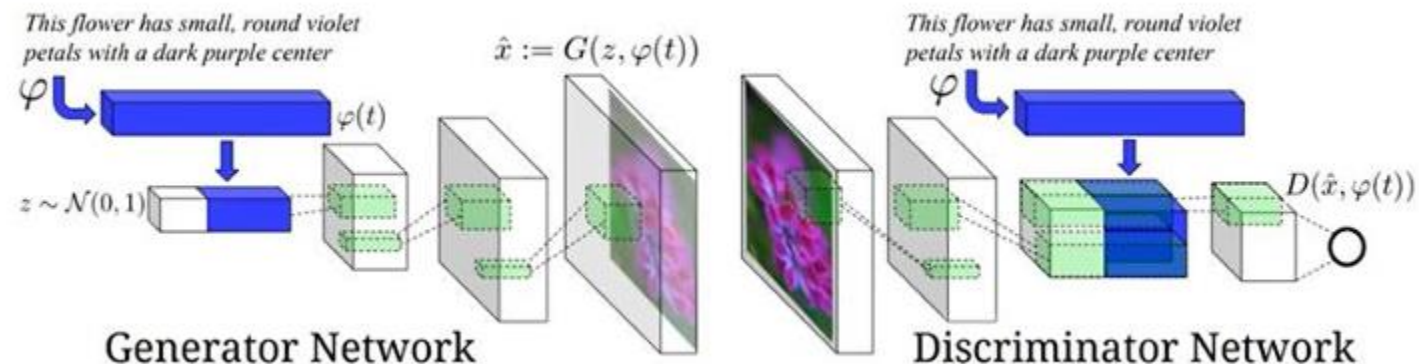
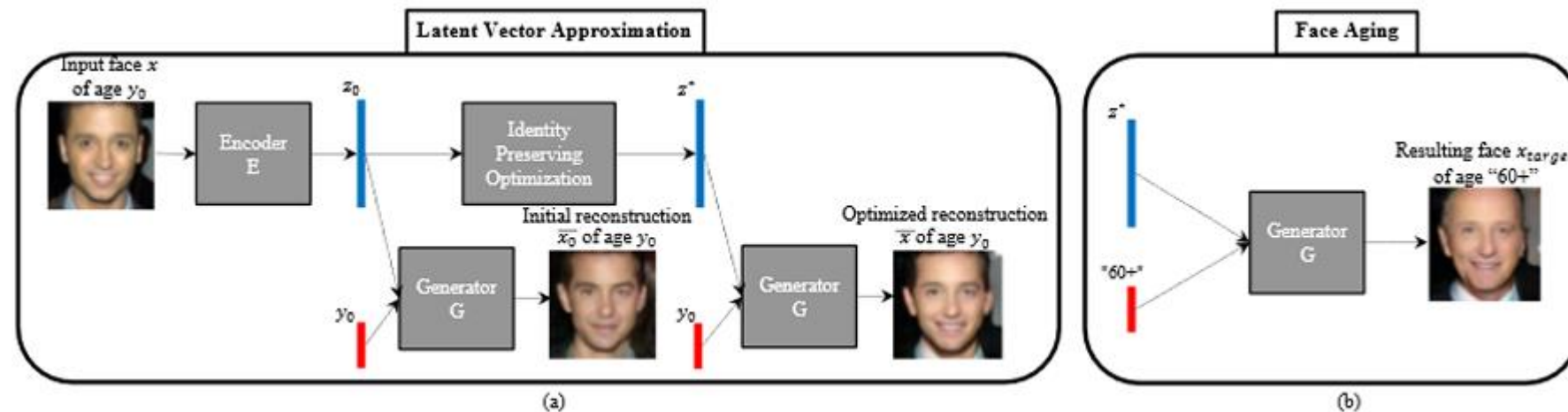


An alternative to *Binary Cross-Entropy (BCE)* loss is **Earth Moving Distance (EMD)** loss: how much ‘work’ is needed to transform one probability distribution into another. This means the loss is not restricted to $[0, 1]$, preventing mode collapse (generator producing the same images). (this is also called **Wasserstein GANs**)

Generative AI - GANs

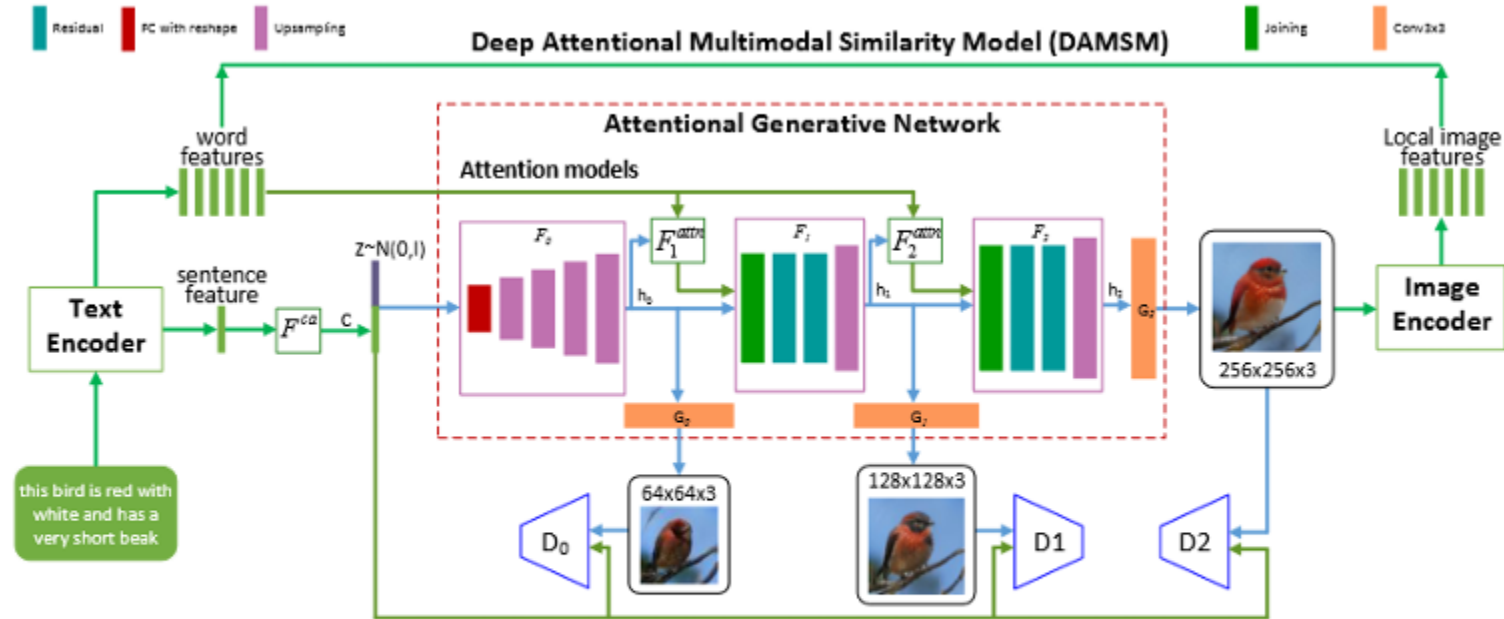
Identity-Preserving Optimization: Minimize the distance between encodings of the original and the reconstructed image.

- **Conditional GANs** allow the generation of images with conditional information (e.g. class labels, age, text description).



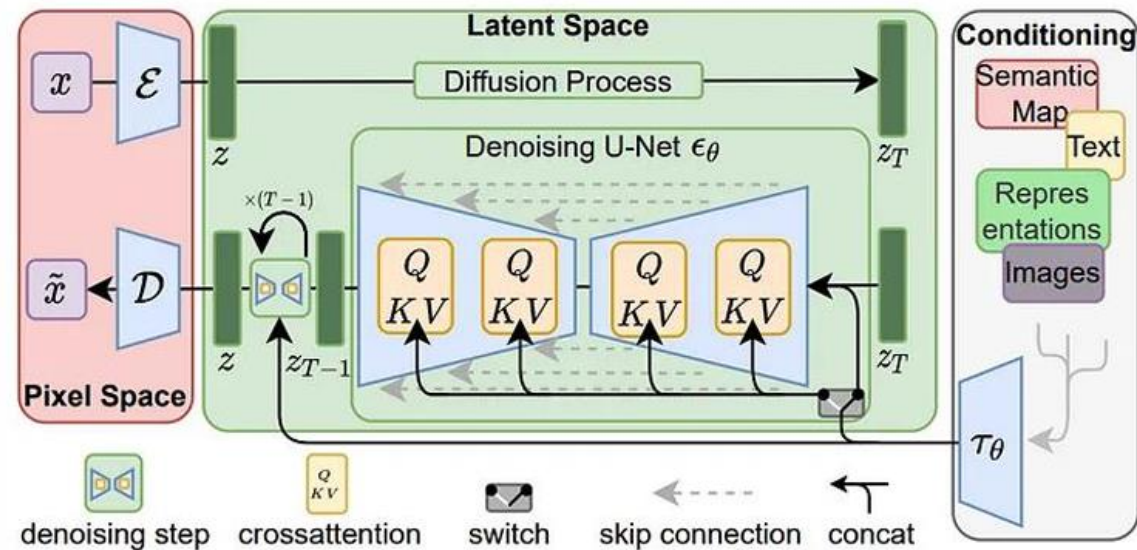
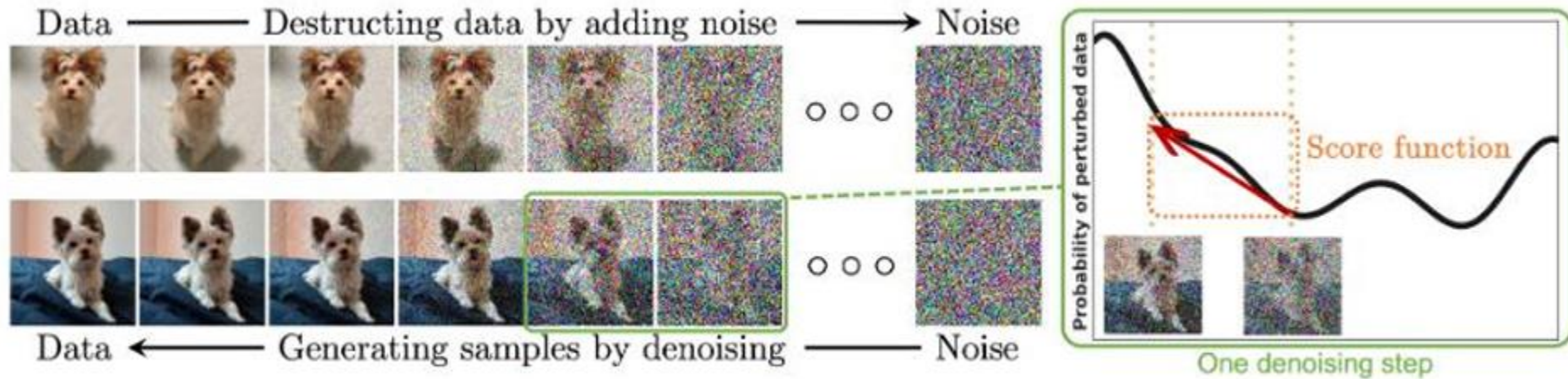
Generative AI - GANs

- **AttnGAN**: fine-grained text to image generation using attention.



$$\mathcal{L}_{G_i} = \underbrace{-\frac{1}{2} \mathbb{E}_{\hat{x}_i \sim p_{G_i}} [\log(D_i(\hat{x}_i))]}_{\text{unconditional loss}} - \underbrace{\frac{1}{2} \mathbb{E}_{\hat{x}_i \sim p_{G_i}} [\log(D_i(\hat{x}_i, \bar{e}))]}_{\text{conditional loss}},$$

Generative AI - Diffusion Models



Generative AI - Diffusion Models

- A popular text to image diffusion model is **DALL-E**:
 1. Train a VQ-VAE to encode each image into a 32x32 grid of possible images, in which each element can assume one of 8192 possible values.
 2. Train a transformer to model the joint distribution of text and image tokens.



(a) a tapir made of accordion. a tapir with the texture of an accordion. (b) an illustration of a baby hedgehog in a christmas sweater walking a dog (c) a neon sign that reads "backprop". a neon sign that reads "backprop". backprop neon sign (d) the exact same cat on the top as a sketch on the bottom